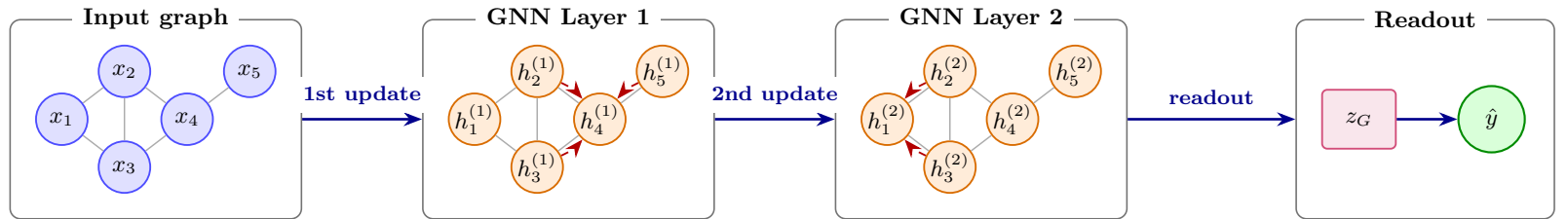


Graph Neural Network: Message Passing Schematic



Node features
 $x_v \in \mathbb{R}^{d_0}$

$$h_v^{(1)} = \sigma \left(\mathbf{W}_{\text{self}}^{(1)} x_v + \sum_{u \in \mathcal{N}(v)} \mathbf{W}_{\text{neigh}}^{(1)} x_u \right) \quad h_v^{(2)} = \sigma \left(\mathbf{W}_{\text{self}}^{(2)} h_v^{(1)} + \sum_{u \in \mathcal{N}(v)} \mathbf{W}_{\text{neigh}}^{(2)} h_u^{(1)} \right)$$

$$z_G = \text{READOUT}(\{h_v^{(2)}\}_{v \in V})$$

$$\hat{y} = f_{\theta}(z_G)$$

Legend:



input feature



hidden state



prediction



graph edge



message